WORKING PAPER

Preventing Environmental-Claim Misrepresentation in the European Union: An Exploratory, Partially Agent-Based Comparison of Stylised Ex-Post Supervision, Voluntary Pre-Screening, and Provenance-Enabled Data Exchange

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Exploratory simulation study. Results are not empirical forecasts, causal estimates, legal advice, budget estimates, or evidence of real-world policy effectiveness.
Model/campaign cut-off: 15 July 2026; legal-status review updated 3 August 2026.

Abstract

Environmental-claim governance is both an information problem and a capacity problem: supervisors observe claims and incomplete evidence rather than firms' model ground-truth environmental states, while preventive review can delay or suppress defensible communication. This paper develops an exploratory, partially agent-based simulation linking corporate environmental claims to consumers, investors, workforces, a limit-order-book market, and capacity-constrained supervision. With ten fixed corporate profiles, pre-specified agent classes, and decision rules defined ex ante, it is a transparent comparison of stipulated mechanisms rather than a search for open-ended, unanticipated emergent behaviour. It compares a stylised ex-post baseline with two hypothetical add-ons: a voluntary rule-based pre-screening hub and a provenance-enabled environmental-data connector. A 28-parameter Latin-hypercube design evaluates 200 configurations at four horizons, using three paired common-random-number replications per configuration; a post hoc outcome-diverse subset of 12 configurations is re-evaluated with 15 replications. In the archived model outputs, the hub has a positive estimated paired effect on exposure-weighted claim severity in 85.0–90.5% of sampled configurations, while increasing the modelled voluntary-communication gap and public and firm costs. The connector's severity pattern is mixed at 120 and 365 days and positive in 97.0% and 100.0% of configurations at 1,000 and 2,000 days, while adding cost and conflict workload. These horizon patterns are descriptive and are confounded with reporting cycles, discounting, and the model's legal calendar. Default rankings reverse across horizons in 80.5% of matched configurations and are sensitive to normative weights. The results are exploratory simulation comparisons, not empirical probabilities, causal estimates, legal conclusions, forecasts, or evidence that any institution would be effective in practice.

Declaration of Generative AI Use

Generative AI tools were used in the development of code and in drafting, writing, and editing the manuscript. Fable 5 was used to support code development, and ChatGPT 5.6 Sol was used to support manuscript drafting and writing. All AI-assisted code, text, citations, and references were reviewed and verified by the author. The simulations were executed computationally, and the reported results were not generated by AI.

Keywords: agent-based modelling; environmental claims; greenwashing; greenhushing; consumer protection; EU environmental law; regulatory technology; algorithmic governance; provenance; uncertainty analysis

1. Introduction

Environmental communication is economically consequential because it can alter consumer choice, investor valuation, employee identification, and access to finance. The same communication is difficult to govern. A claim can be inaccurate because management strategically exaggerates, because the evidence is incomplete, because operational and reporting boundaries are not comparable, or because uncertainty is poorly disclosed. A public authority rarely observes the firm's physical environmental state directly. It works through claims, records, assurance procedures, complaints, public registers, and finite investigative capacity. The regulatory problem is therefore not simply to classify statements as true or false. It is to design procedures that improve the reliability of communication without confusing measurement error with misconduct or inducing firms to withdraw defensible information.

Research has described institutional, market, organisational, and individual drivers of misleading environmental communication (Delmas and Burbano, 2011), the diversity of practices grouped under greenwashing (Lyon and Montgomery, 2015), and the role of accusation and legitimacy in claim interpretation (Seele and Gatti, 2017). A theoretical model shows that stronger scrutiny can deter exaggerated disclosure while reducing disclosure by some firms (Lyon and Maxwell, 2011). An experiment in a specified sustainability-rating context reports a deterrence effect (Parguel, Benoit-Moreau, and Larceneux, 2011); it does not establish a general effect of third-party ratings. Selective disclosure depends on the claim, omitted information, audience, and surrounding evidence (Marquis, Toffel, and Zhou, 2016).

Greenhushing and strategic silence have been documented in particular tourism and certification settings (Font, Elgammal, and Lamond, 2017; Carlos and Lewis, 2018), without establishing their prevalence across EU environmental claims.

These observations motivate a computational approach. Agent-based computational economics represents economic processes as dynamic systems generated by heterogeneous, interacting agents (Tesfatsion, 2006). Agent-based finance is especially useful when investor strategies, market microstructure, credit, and feedback between prices and balance sheets matter (LeBaron, 2006). In the present model, corporate environmental investment and communication interact with heterogeneous consumers, fundamentalist and chartist traders, employees, a bank, a central bank, the State, assurance providers, and a capacity-constrained supervisor. The resulting system is not an empirical digital twin. It is a transparent computational laboratory in which institutional mechanisms can be altered while paired economic shocks are held constant.

The research question is: under which sampled model configurations do a stylised ex-post supervision baseline, a voluntary rule-based pre-screening add-on, and a provenance-enabled data-exchange add-on produce different claim-quality, communication, workload, and cost outcomes? The comparison is conditional and does not identify a universally optimal institution or estimate legal effectiveness.

The analysis isolates two hypothetical add-on mechanisms. Preventive screening can correct a draft but can also delay or suppress defensible communication. Provenance-enabled exchange can alter available evidence only as authorisation, transfer, reconciliation, and correction accumulate. Actor-specific information constraints make false positives, inconclusive cases, evidence conflicts, and the modelled voluntary-communication gap endogenous to procedure, subject to the metric-construction limits described below.

The principal simulation and legal-parameter cut-off is 15 July 2026; legal status in this revision is updated to 3 August 2026. The model distinguishes binding instruments, national-transposition conventions, stylised legal proxies, and hypothetical institutions. The pre-screening hub and provenance-enabled connector are experimental add-ons, not EU obligations. COM(2023)166 is not baseline law; its pre-verification switch is disabled.

The remainder of the paper describes the institutional map, model architecture, experimental design, findings, contributions, limitations, and archival provenance package.

2. Institutional and Regulatory Context

2.1 A layered rather than unitary legal map

The model does not posit a single EU "greenwashing code." It instead assigns claims to distinct tracks whose scope, dates, evidence requirements, and remedies differ. This separation is substantively important. A consumer-facing environmental advertisement, a mandatory sustainability report, an investor communication, a Taxonomy alignment statement, and a due-diligence failure do not share the same legal basis or penalty ceiling.

Table 1. Calendar-aware legal and institutional map

Model/campaign legal-parameter cut-off: 15 July 2026; legal-status review updated 3 August 2026. Experimental institutions are not stated as EU obligations.

Track	Model use	Date/default	Classification
UCPD	Misleading actions and omissions in B2C claims	Active throughout	Binding-law representation

Track	Model use	Date/default	Classification
Directive 2024/825	Generic claims, labels, offsets, future performance	Applies 27 Sep 2026	Binding-law representation
CSRD / Directive 2026/470	Scope, structured reporting, limited assurance	Scenario: 19 Mar 2027	Law plus configurable transposition
EU Taxonomy	Claim-triggered consistency proxy; not Article 8 scope	When modelled Taxonomy claim is made	Stylised legal proxy
Issuer communications	MAR, prospectus, transparency screens	Scope dependent	Stylised screening proxies requiring specialist review
CSDDD	Due-diligence date/remedy gate; full undertaking scope not implemented	Applies 26 Jul 2029	Stylised due-diligence proxy requiring specialist review
European Green Bonds	Voluntary designation and use of proceeds	Instrument specific	Binding-law representation
Green Claims pre-verification	Disabled counterfactual	Off	Experiment, not baseline law

The UCPD supplies the general business-to-consumer baseline for misleading actions and omissions and allows substantiating evidence to be required (European Parliament and Council, 2005). Directive (EU) 2024/825 introduced selected Annex I prohibitions concerning specified environmental practices (European Parliament and Council, 2024a). Member States were required to transpose it by 27 March 2026 and apply national measures from 27 September 2026. The model represents selected B2C Annex I prohibitions from that date; it does not treat every environmental or future-performance claim as a per se infringement.

Directive (EU) 2019/2161 strengthens enforcement and penalty availability for widespread infringements and widespread infringements with a Union dimension (European Parliament and Council, 2019). It requires Member States to set the maximum fine for the relevant infringements at no less than 4% of the trader's annual turnover in the Member State or Member States concerned. It does not establish 4% as a uniform EU ceiling. The unchanged model therefore treats an exact 4% cap as a LEGAL-ANCHOR scenario choice, available only when a case is classified as a coordinated widespread cross-border consumer infringement. Ordinary simulated cases use a separate 1% experimental cap applied to a simulation-scale turnover proxy.

The corporate-reporting proxy draws on Directive (EU) 2022/2464 as amended by Directive (EU) 2026/470 (European Parliament and Council, 2022a; 2026). The amended principal threshold requires net turnover exceeding EUR 450 million and average employment exceeding 1,000 for financial years starting on or after 1 January 2027; both conditions must be met. The model's 19 March 2027 switch is an experimental transposition convention, not a uniform substantive application date. It uses selected concepts from the 2023 ESRS E1 and does not implement the revised ESRS adopted by the Commission on 3 July 2026 while those standards remained under scrutiny at this revision date (European Commission, 2026).

The model distinguishes Taxonomy eligibility from alignment and separately records turnover, capital expenditure, and operating expenditure (European Parliament and Council, 2020). Its claim-triggered mechanism is a reduced-form consistency screen, not an implementation of Article 8 reporting scope, technical screening criteria, minimum safeguards, or current templates. The connector does not populate Taxonomy or net-zero fields.

Issuer and capital-market communications are screened by stylised proxies inspired by the Market Abuse Regulation, Prospectus Regulation, and Transparency Directive (European

Parliament and Council, 2014; 2017; 2004). These proxies are not doctrinal implementations. The UCPD is not used as a universal basis for investor-directed communication. A voluntary European Green Bond-style accounting track preserves the model identity between issuance, unspent proceeds, and green expenditure but does not claim full compliance with Regulation (EU) 2023/2631.

The CSDDD is represented only as a reduced-form due-diligence procedure (European Parliament and Council, 2024b, as amended in 2026). Its 3% maximum is callable only for due-diligence cases, and the model's application date is 26 July 2029. Directive (EU) 2026/470 also revises the principal undertaking threshold to more than 5,000 employees and more than EUR 1.5 billion net worldwide turnover, with separate group, franchise, and third-country rules. The current model gates CSDDD by date and remedy track but does not implement that complete firm-scope test; it must therefore not be described as a doctrinally complete CSDDD implementation. Greenhushing is not modelled as an offence.

2.2 Experimental institutions

Arm B is a hypothetical voluntary, non-binding, rule-based drafting and feedback service for model-eligible undertakings outside the modelled reporting scope; it is not an enacted pre-verification obligation and the 1,000-employee line is not an SME definition. Drafts may be revised, withdrawn, or published after review, with no endorsement. Deployment would require a separate AI Act assessment based on intended purpose, provider/deployer roles, architecture, and official use. The paper does not classify the hub as high-risk or exempt, and no adverse legal determination is made solely by it (European Parliament and Council, 2024d).

Arm C is a hypothetical provenance-enabled interoperability and evidence-reconciliation layer, not a sensor network, Digital Product Passport, certification system, or enacted reporting obligation. GHG accounting (GHG Protocol, 2004; 2011), ESRS E1 (European Commission, 2023b), the Data Act (European Parliament and Council, 2023b), ESPR (European Parliament and Council, 2024c), and PACT (Partnership for Carbon Transparency, 2025a; 2025b) supply limited design analogies, not legal authority for access or reuse. Implementation would require source-specific authority, contracts or licences, correction and retention rules, trade-secret safeguards (European Parliament and Council, 2016a), data-protection compliance where applicable (European Parliament and Council, 2016b), and cybersecurity governance (European Parliament and Council, 2022b). A connector record may initiate modelled reconciliation but cannot sanction a firm by itself.

COM(2023)166 remains formally pending; at the legal-status review date, the European Parliament’s Legislative Observatory recorded the file as awaiting the Council’s first-reading position (European Commission, 2023a; European Parliament, n.d.). It is used only to delimit counterfactual space; the model’s pre-verification switch is off. No result evaluates an enacted Green Claims Directive.

2.3 Related work and positioning

The greenwashing literature motivates the model's distinction among overstatement, incomplete evidence, selective disclosure, and communication withdrawal. Audit threat can reduce exaggerated disclosure while discouraging some truthful disclosure (Lyon and Maxwell, 2011); scrutiny and organisational norms shape selective disclosure (Marquis, Toffel, and Zhou, 2016); and greenhushing or strategic silence can limit favourable communication (Font, Elgammal, and Lamond, 2017; Carlos and Lewis, 2018). These mechanisms support endogenous revision and withdrawal, but they do not empirically calibrate either experimental institution.

RegTech and SupTech research emphasises automated data collection and screening, prospective surveillance, supervisory capacity, data quality, explainability, and operational risk (Broeders and Prenio, 2018; Financial Stability Board, 2020). Corporate GHG standards (GHG Protocol, 2004; 2011), ESRS E1 (European Commission, 2023b), the Data Act (European Parliament and Council, 2023b), Digital Product Passport architecture under ESPR (European Parliament and Council, 2024c), and PACT specifications (Partnership for Carbon Transparency, 2025a; 2025b) provide closer foundations for accounting boundaries, provenance, identifiers, and interoperable value-chain data exchange. They motivate design dimensions of the connector rather than validate its parameters or make it an implementation of an existing obligation.

Financial agent-based models are commonly assessed against fat-tailed returns, volatility clustering, and limited linear return predictability (Cont, 2001; Lux and Marchesi, 1999). Reproducing selected stylised facts is weaker than empirical calibration or out-of-sample validation. This paper therefore reports each diagnostic as reproduced, partially reproduced, or not reproduced.

Against these literatures, the comparison asks how preventive screening, interoperable evidence, and ex-post supervision differ when evidence remains uncertain and administrative capacity is scarce. The contribution is the joint mechanism test under common shocks, not a claim that the individual components are novel or that the experiment estimates policy effects.

3. Model and Methodology

3.1 Purpose, entities, and scales

The model is a daily, discrete-time, non-spatial agent-based policy simulation with an embedded limit-order book. Day 1 maps to 1 January 2026. The headline campaign uses ten fixed corporate profiles, eight traders plus market makers, four aggregate consumer segments, no manipulators, and disabled credit. Trading occurs on every simulated day; the model does not distinguish weekends from business days. The description is ODD-inspired rather than a complete ODD protocol (Grimm et al., 2020).

This is a deliberately partial use of agent-based modelling. The policy campaign contains ten fixed corporate profiles, a small and pre-specified set of agent classes, and decision rules and institutional procedures defined ex ante. Random shocks and interactions generate variation in simulated trajectories, but the model is not designed to produce open-ended adaptation or unforeseen forms of collective behaviour. Accordingly, the analysis is a comparison of stipulated mechanisms: any pattern that emerges is conditional on the specified agents, profiles, rules, and parameter ranges, rather than a discovery claim about unanticipated real-world dynamics.

Firms differ in financial state, environmental state, workforce, communication profile, and evidence. The policy campaign contains fundamentalist, chartist, noise, and market-making behaviour; manipulation and credit modules exist in the codebase but are disabled in the reported policy runs. Consumers differ in attention, environmental preference, sophistication, and memory. Employees respond to modelled internal signals and published supervisory outcomes. State, assurance, supervision, dividends, and trading frictions remain active as specified in the campaign manifests.

3.2 Information-safe architecture

INFORMATION-SAFE STYLIZATION: the architecture separates the physical or latent environmental state from the information available to every operational decision-maker. This boundary is substantive: claim accuracy and supervisory performance must arise from evidence and procedure rather than oracle access.

Table 2. Information boundaries by actor

Latent truth is accessible only to the ex-post research evaluator.

Actor	Latent facts	Accessible evidence/information	Decision output
Firm	Own operations	Own records, public information	Investment, evidence, claim, qualification
Assurance provider	None	Mandatory report and submitted records	Uncertain assurance estimate/opinion
Supervisor	None	Claims, requested records, complaints, assurance, connector data	Screening, case, remedy, publication
Consumer	None	Attended claims, noisy linked evidence, decisions	Demand allocation and belief update
Investor	None	Environmental context, linked evidence by sophistication	Orders and environmental fair value
Employee	None	Noisy internal signal and public outcomes	Trust, productivity, turnover
Research evaluator	Ex-post snapshot only	Truth, intent, claims, evidence, outcomes	Research-only performance metrics

Each firm observes its own operations and may create claims and evidence. An assurance provider receives the mandatory report and the relevant records, re-performs uncertain procedures, and produces an opinion without seeing latent truth. A supervisor receives claims, submitted or requested evidence, assurance outputs, complaints, whistleblower signals, connector records, and prior cases. Consumers observe attended public claims, linked public evidence with noise, and published decisions. Investors receive an environmental context rather than a latent score; sophisticated agents can inspect linked evidence more fully than unsophisticated agents. Employees receive a noisy internal operational signal and public outcomes. Only the research evaluator accesses the private claim-to-truth mapping after the simulation path has completed. Tests in the repository are designed to prevent that mapping from entering agent decisions.

This separation resolves a common simulation problem: a regulator with direct access to truth would mechanically outperform realistic institutions, while consumers or investors using perfect information would eliminate the very asymmetry under study. Here, supervisory accuracy is an outcome of procedure and evidence, not an assumed oracle.

3.3 Environmental state, claims, and evidence

The environmental fact vector records Scope 1, 2, and 3 emissions; energy use and renewable share; water; waste and recycling; pollution; biodiversity pressure; Taxonomy-eligible and aligned turnover, capital expenditure, and operating expenditure; real environmental capital expenditure; and offsets separated from gross emissions. Records carry periods, group or operational boundaries, methodologies, and uncertainty. Scope categories and boundary metadata are motivated by GHG Protocol and ESRS E1 reporting architecture, but the vector is a STYLIZATION, not a complete compliant inventory or lifecycle assessment (GHG Protocol, 2004, 2011; European Commission, 2023b).

An environmental claim is a structured object rather than a scalar message. It identifies the firm, date, channel, audience, claim type, metric, asserted value, unit, period, boundary, linked evidence, uncertainty, qualifications, target date, offset reliance, and correction or withdrawal status. Because enforcement compares named quantities, the firm's aggregate green score is retained only for compatibility and research dashboards; it is not the sanctioning basis.

Evidence records include source, subject, period, estimate, standard error, coverage, independence, verification status, and confidence. Management evidence is neither silently converted into independent evidence nor treated as certain. The connector and limited-assurance services produce their own records with strictly positive uncertainty. The supervisor chooses relevant evidence by subject, coverage, independence, period, and accessibility.

For a metric where a larger value is environmentally better, the evidence divergence is:

$$d = \text{asserted value - evidence estimate},$$

and the one-sided standardised divergence is:

$$z\_over = \max(0, d) / \text{sqrt}(\text{evidence error}^2 + \text{stated uncertainty}^2 + \text{positive floor}).$$

The sign reverses for metrics where a lower value is greener, including emissions, pollution or water intensity, biodiversity pressure, and net-emission ratios. Materiality is the greener-than-evidence gap divided by a unit-aware scale. Default experimental bands classify z below 1 as noise; z from 1 to below 2.5 as inconclusive or requiring more evidence; statistically visible but sub-2% divergence as a correctable error; material qualified divergence as negligence; stronger material divergence as overstatement; and repeated severe findings as systemic abuse. Applicable per se rules after 27 September 2026 can override the statistical band.

The detector includes specific safeguards. Self-declared uncertainty is plausible only up to twice the evidence standard error; excess uncertainty is disregarded and flagged. Restricted boundaries do not widen the error band unless meaningfully qualified. Repeated one-sided sub-threshold findings can escalate only after at least four findings within 365 days with a mean z-score of at least 0.5. Conflicting independent and firm records route to an inconclusive conflict process rather than a sanction. Cross-period checks address cherry-picked periods and implausible baselines. Placeholder qualifications do not count as meaningful.

3.4 Corporate choice and the modelled voluntary-communication gap

CORPORATE-CHOICE STYLIZATION: at reporting epochs, the firm searches a transparent discrete grid over real environmental investment, voluntary communication intensity, qualification, evidence effort, and possible overstatement. The objective contains a green communication premium, missed truthful premium, investment and evidence costs, qualification cost, compliance burden, and expected enforcement. The discrete grid keeps the mechanism inspectable; its coefficients and choice responses are not behaviourally estimated.

The truthful communication benchmark is:

$$q\_truthful = \text{clip}(0.20 + 0.70 \times \text{real environmental score}, 0, 1),$$

and the modelled voluntary-communication gap is:

$$\text{modelled voluntary-communication gap} = \max(0, q\_truthful - q\_chosen).$$

VOLUNTARY-COMMUNICATION-GAP STYLIZATION: the modelled voluntary-communication gap compares the communication that the firm’s current environmental score could truthfully support with the voluntary communication it chooses. It captures suppression relative to an internally available truthful benchmark, not an observed legal violation, observed managerial behaviour, or a claim about managerial intent. Mandatory reporting remains separate. A firm can therefore reduce voluntary truthful communication without violating a reporting obligation;

that choice eliminates the relevant voluntary claim, so the model does not invent a sanction for silence, but it reduces visibility and the truthful green premium.

The hub can alter this choice by imposing a review delay, producing warnings, encouraging meaningful revision, or prompting withdrawal. The model distinguishes prevention of good-faith errors from withdrawal-induced suppression. It reports intention-to-treat participation and treatment-on-the-treated revision measures, while corporate strategy labels remain research-only.

3.5 Assurance, supervision, and remedies

Every claim receives low-cost screening. Default experimental capacity permits 20 evidence requests and five formal investigations per reporting period, with 10% random surveillance. Complaints and whistleblowers receive higher priority; channel, claim type, evidence gap, and history contribute to the remaining risk score. Random surveillance reduces the ability of firms to game a deterministic ranking.

Cases move through recorded states: screened; evidence requested; under assessment; correction window or formal investigation; decided; published; and closed. Noise and inconclusive cases may close early. First-time correctable errors receive an experimental 30-day window, with modelled compliance at the midpoint. No monetary transfer occurs before a decision.

The ordinary simulated sanction calculation is:

$$\textit{calculated sanction} = 1.5 \times \textit{estimated benefit} + 0.02 \times \textit{affected revenue} \times \textit{severity} \times \textit{confidence}.$$

Recidivism can increase the calculated amount, after which the applied sanction is the minimum of the calculated amount, the track-specific ceiling, and corporate headroom. Simulation-scale turnover equals an experimental multiple of corporate balance; statutory annual net turnover is kept separate for legal scope tests. The 4% consumer and 3% due-diligence ceilings are track-gated. Qualification, correction, withdrawal, publication, redress, and sanctions are separate remedies.

Corrections are prospective and immutable. The original asserted value is preserved, a correction event records exposure days and case provenance, and incidence metrics remain based on the original claim. Consequently, a later correction cannot erase the historical period of misleading exposure.

3.6 Conflict resolution and the connector

Evidence disagreement is treated procedurally. A material conflict opens a CONFLICT_RESOLUTION case, enters the shared investigation queue at priority 0.85, and consumes capacity when opened. A default 20% capacity reserve prevents the conflict desk from being starved by higher-priority enforcement matters, but reduces capacity available to ordinary cases. Both quantities are experimental.

An investigation commissions a fresh measurement. Under the connector regime, the register's correction lifecycle can produce a superseding record after a delay. Otherwise a stylised third party re-measures after 30 days. The supervisor sees only the new uncertain evidence record, not physical truth. Agreement with the firm confirms the claim and corrects the external record; agreement with the register permits escalation through the ordinary evidentiary path; three-way disagreement leads to dismissal; and timeout permits confirmation, correction demand, or dismissal. Escalation without corroborated evidence is impossible by construction.

The connector authorises each firm-source pair independently. Eight default source types cover Scope 1 and 2, renewable share, water, waste and recycling, pollution, partial Scope 3, and environmental capital expenditure. The model assumes that an underlying source record exists and operates at firm-source level rather than representing every facility, machine, subsidiary, supplier, or service-production boundary. Provenance is hashed over the payload, methodology version, period, uncertainty, completeness, and correction status, and hashes are verified before use. Data can be stale, incomplete, mismatched, unavailable, or wrong; coverage widens effective uncertainty. Reconciliation classifies matched data, rounding noise, methodological difference, incomplete coverage, source conflict, correction required, suspicious override, material overstatement, or repeated manipulation. Reconciliation itself never imposes a sanction.

3.7 Consumers, investors, employees, and financial feedback

Consumers allocate an exogenous EUR 1,000 daily budget across firms through logit utility. Segment-level environmental preferences, attention, sophistication, and memory determine the effect of claims and evidence. Accessible claim-evidence divergence reduces utility. A larger modelled voluntary-communication gap reduces visibility and entails a foregone truthful premium but does not create a controversy signal. The budget is an explicit external injection: daily gross product revenue equals the configured budget, and gross revenue is reconciled with production cost and corporate margin.

Fundamentalist investors receive an environmental context, not truth. Their environmental fair value is:

$$V\_fair = V\_fundamental \text{ x } (1 + \text{greenium gamma x posterior score x credibility}) \text{ x } (1 - \text{controversy discount}).$$

Sophisticated investors inspect linked evidence and respond more quickly to controversy; unsophisticated investors update more slowly. Chartists retain an indirect price-only channel. Market makers supply liquidity, other trader types place orders, and the order book determines transaction prices. Credit, collateral, margin calls, liquidation, interest, dividends, and bank exposure propagate environmental communication into financial outcomes without directly inserting the latent environmental vector into trading rules.

Claims, accessible evidence, and public outcomes affect investor inputs, order flow, and prices. A limited state-mediated financing path also exists: a corporate policy may sell treasury shares using the own order-book midpoint, sweep the proceeds into the corporate balance, and later use that balance in transition-spending, reporting-affordability, penalty, preparation, and evidence-cost calculations. A static audit detected no explicit price read inside the focal communication/transition decision methods, but it cannot prove the absence or immateriality of indirect feedback. No causal claim about equity-price discipline is made; a dynamic treasury-path ablation is required before materiality can be assessed.

Employees update trust from an evidence-accessible internal discrepancy and from published decisions. Quiet truthful periods allow slower recovery. Productivity affects product cost once:

$$\text{productivity multiplier} = 1 - 0.03 \text{ x } (1 - \text{trust}).$$

Annual workforce turnover begins at 10%, is capped at 30%, and is converted to a daily hazard. Departures, vacancies, hiring, 30-day onboarding, and replacement costs feed corporate accounts. The rolling 365-day employee average subsequently enters the reporting-scope test.

3.8 Event order, random streams, and accounting invariants

On each active market day the simulation: maps the legal date; updates fundamentals and environmental transition; processes prior public information through the workforce; runs product demand and books revenue and cost; creates scheduled reports and voluntary claims; performs limited assurance; screens claims and advances cases; forms information-safe investor and banking inputs; runs trading, credit, interest, dividends, and evolutionary review; and logs and validates ledgers.

This schedule prevents a claim created during the current day from affecting that day's earlier consumer demand. Newly published claims and investor-accessible linked evidence may nevertheless affect same-day trading; a public supervisory decision is an additional, not necessary, information channel. Separate pseudo-random streams are used for several processes, but regime-dependent event counts may shift sequential stream consumption after arms diverge.

Accounting checks enforce single-entry pathways for corporate margin, productivity, sanctions, subsidies, evidence and replacement costs, and green-bond proceeds. A sanction received by the State equals the amount removed from corporate balances. Controversy changes demand or valuation but does not reduce the same fundamental twice.

3.9 Policy regimes

Table 3. Policy regimes compared

Arm	Institution	Primary mechanism	Legal status
A	Stylised ex-post baseline	Capacity-constrained screening, evidence requests, assurance, investigation, correction, and sanctions	Composite stylisation of selected legal mechanisms
B	Voluntary rule-based pre-screening hub	Non-binding draft feedback, revision, withdrawal, and delayed publication	Hypothetical add-on; not an enacted obligation
C	Provenance-enabled environmental-data connector	Authorised transfer, provenance, reconciliation, correction, and conflict handling	Hypothetical add-on; not a certification regime

Archived identifiers: A = current_eu_supervision; B = sme_algorithmic_prescreening; C = certified_green_data_connector. The legacy terms 'SME' and 'certified' are retained only for traceability: the 1,000-employee line is not an SME definition, and Arm C is not a certification regime.

Arm A preserves the stylised supervision institution and legal calendar. Arm B adds voluntary pre-publication feedback; Arm C adds provenance-enabled transfer, reconciliation, and register correction. The contrasts therefore compare A with A+B and A with A+C under shared initial seeds; they do not compare coequal legal systems or identify B×C complementarity. Uptake and authorisation are model-generated, type-dependent Bernoulli processes based on fixed profiles, not strategic endogenous selection. Participant-conditioned quantities remain descriptive.

4. Experimental Design and Reproducibility

4.1 Paired comparison

Within each replication, all three regimes receive identical initial firms, traders, populations, strategies, latent environmental trajectories, macro shocks, market seed, and supervision seed.

Only the State intervention mechanism and dedicated policy streams differ. This common-random-number design reduces irrelevant between-arm variation. Across replications and parameter draws, the stochastic environment changes.

For draw j and replication r, the archived campaign uses:

$$market\ seed = 20{,}260{,}716 + 1{,}000{,}000\ x\ j + 1{,}000\ x\ r,$$

$$supervision\ seed = 104{,}729 + 104{,}729\ x\ j + 7{,}919\ x\ r.$$

The runner explicitly re-seeds each regime, making results invariant to whether arms are executed A-B-C, C-B-A, or separately. The repository contains tests for same-seed reproducibility, order invariance, no truth leakage, correction immutability, sanction monotonicity and track gating, conservation identities, connector provenance, conflict corroboration, campaign resume behaviour, and the disabled-feature legacy trajectory.

4.2 Global sensitivity campaign

Latin-hypercube sampling stratifies each parameter dimension across its range and is a standard design for computer experiments (McKay, Beckman, and Conover, 1979). The campaign samples 28 parameters classified as EXPERIMENT or STYLIZATION. Legal dates and statutory ceilings are fixed. Sampled families include the sanction-scale bridge, evidence and investigation capacity, surveillance, regulatory strictness, conflict-desk parameters, hub participation and error, connector coverage and failures, consumer response, investor sophistication, workforce trust, compliance burden, and the social discount rate.

Table 4. Attached global sensitivity campaign

Each regime run is paired within a draw-replication cell. The archived configuration records no subsets.

Horizon	LHS draws	Paired reps	Regimes	Runs	Subset
120 days	200	3	3	1,800	None
365 days	200	3	3	1,800	None
1,000 days	200	3	3	1,800	None
2,000 days	200	3	3	1,800	None
Total	800 horizon-draws	-	-	7,200	Full four-horizon campaign

Each horizon contains 200 parameter draws, three paired replications, and three arms: 1,800 arm-runs per horizon and 7,200 in total. The top-level configuration and per-horizon manifests jointly record the campaign design and execution identity; configuration.json alone does not contain the code version, realised 200×28 design matrix, credit setting, or manipulator count.

The machine-readable configuration in results/configuration.json records the authoritative four-horizon design: all four horizons have subset: null, and each summary contains 200 draws.

4.3 Outcomes and inference

The evaluator distinguishes research-only truth-based metrics from quantities observable to in-model institutions. Research-only outcomes include original material overstatements, exposure-weighted severity, population detection recall, severity-weighted recall, strategic versus measurement-noise incidence, and withheld truthful claims. They are calculated after a run and never supplied to agents.

Observable or procedural outcomes include backlog, queue age, case completion, conflict cases, correction delay, hub participation, meaningful revisions, noise flags, the modelled voluntary-communication gap, public and firm cost, employee trust, replacement cost, investment, and final emissions. Original incidence is based on claims as published; live-uncorrected incidence recognises subsequent correction; exposure-weighted severity multiplies harm by exposure duration; misleading-claim days preserve the temporal burden of delayed action.

For each configuration, paired mean differences are calculated across three replications. The resulting 200 configuration-level effects are summarised. For harmful outcomes and costs, signs are recoded so that a positive value denotes an estimated improvement relative to the stylised baseline. Reported 1.96×SD/√200 half-widths describe dispersion of selected design-space means; they are not empirical confidence intervals. The share improved is the share of sampled configurations whose estimated paired mean has the favourable sign, not a success probability.

Composite rankings use five discrete experimental weight scenarios: default, claim-accuracy priority, communication protection, cost priority, and burden protection. Within each configuration, every criterion is min–max normalised across the three arms before weighting. The score is a relative, alternative-set-dependent ranking device, not a welfare function; the weights are normative assumptions (OECD and European Commission, Joint Research Centre, 2008). Raw metrics and Pareto co-membership are reported first. Standardised coefficients and PRCCs are exploratory associations and can miss ties, interactions, and nonlinearity (Marino et al., 2008; Pianosi et al., 2016).

4.4 Replication robustness extension

Three paired replications per LHS configuration provide an exploratory screen but can under-sample rare escalation, institutional failure, and tail outcomes. After the principal campaign, 12 configurations were selected deterministically by maximin coverage of standardised outcome space and extended to 15 paired replications. This post hoc outcome-diverse subset is not probability-representative of the parameter design.

The extension has been executed. Draws 41, 42, 69, 71, 75, 104, 109, 136, 144, 147, 180, and 187 were extended to 15 paired replications at each horizon, and all 48 draw-horizon extension files pass validation. Moving from three to 15 replications narrows the paired 95% interval in 92.3% of the 1,743 draw-metric-policy cells with a finite comparison, with a median half-width ratio of 0.26. The default-weight winner computed at three replications is confirmed at 15 replications in 32 of 48 draw-horizon cells (9, 9, 8, and 6 of 12 at 120, 365, 1,000, and 2,000 days), and the complete three-regime ranking is confirmed in 60.4% of cells. Draw-level winner labels under three replications are therefore noisy, especially at long horizons; the paper accordingly emphasises distributional effects, improvement fractions, and Pareto membership across the 200 draws rather than draw-level league tables. The first three replications remain the authoritative campaign, and extension outputs are stored separately in results/replication_robustness/.

5. Findings

5.1 Conditional policy maps

Ranking instability explains why the two policy mechanisms must be reported conditionally rather than collapsed into a universal league table. The default-weight winner changes across horizons for 161 of 200 matched parameter draws (80.5%). Within a horizon, changing normative weights changes the winner in 70.0% of 120-day draws, 63.0% of 365-day draws, 58.0% of 1,000-day draws, and 45.5% of 2,000-day draws. These reversals are consequences of distinct accuracy, the modelled voluntary-communication gap, cost, and timing trade-offs, not evidence that comparison is impossible.

Table 5. Default-weight winner shares and within-horizon weight sensitivity

Shares are counts divided by 200 LHS draws. They are not empirical success probabilities.

Horizon	Stylised baseline	Pre-screening hub	Data connector	Weight-sensitive draws
120	47.5%	48.0%	4.5%	70.0%
365	53.0%	38.0%	9.0%	63.0%
1,000	35.5%	13.5%	51.0%	58.0%
2,000	33.5%	7.0%	59.5%	45.5%

At 120 days, the stylised baseline and the hub are nearly tied under default weights, winning 47.5% and 48.0% of draws; the connector wins 4.5%. At 365 days, the stylised baseline leads at 53.0%, followed by the hub at 38.0% and the connector at 9.0%. The ordering changes at long horizons: the connector wins 51.0% at 1,000 days and 59.5% at 2,000 days, while the hub falls to 13.5% and 7.0%. The hub does not cease to improve claim-quality metrics; it loses default-weight competitiveness because review and revision costs recur, participation is selective, and the modelled voluntary-communication gap is penalised, while connector evidence accumulates. The stylised baseline remains competitive because cost and the modelled voluntary-communication gap enter the composite. These frequencies are not probabilities that a policy will succeed in the European economy. They are shares of the model's 200 stratified parameter configurations.

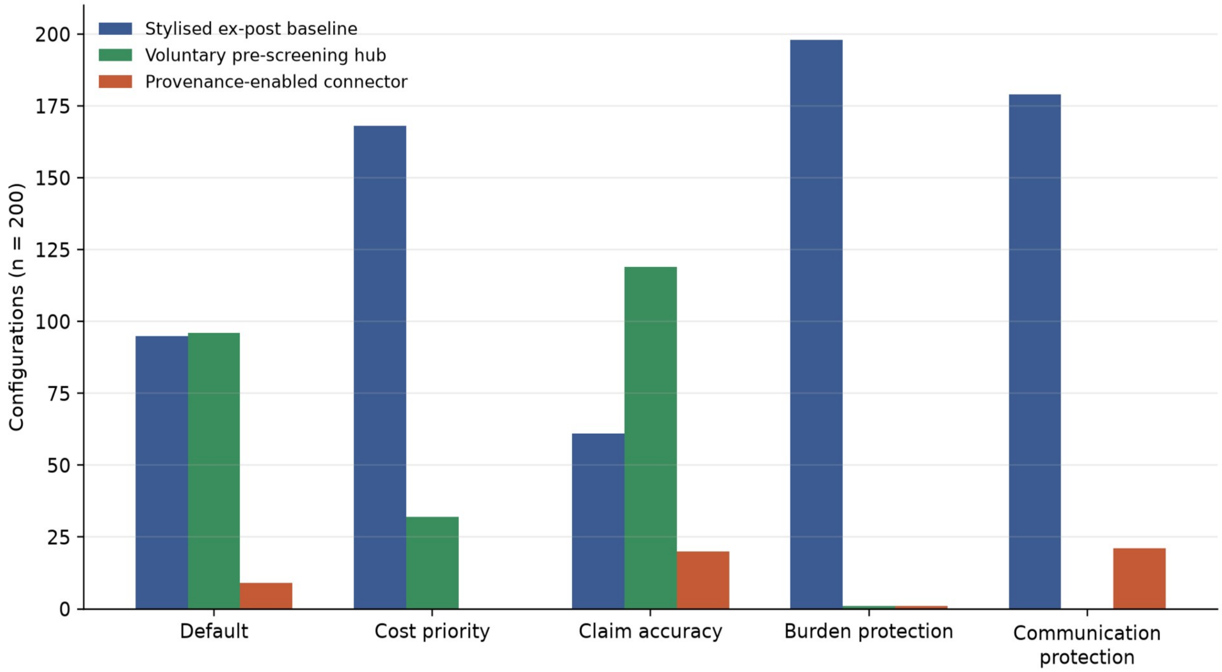

Figure 1. Most frequent arm under five normative weight scenarios at 120 days.

Source: results/summaries/horizon_120d_draw_effects.csv (archived campaign code identifier 1645f3589408503894c7e8f44a92ffe382a9a5b4+dirty). Counts cover 200 configurations with three paired replications; they are not empirical frequencies.

The five weight scenarios explain why no regime dominates. At 120 days, the accuracy-focused scenario selects the hub in 119 of 200 draws, while the cost-focused and communication-protection scenarios select the stylised baseline in 168 and 179 draws. Under the burden-protective weights, the stylised baseline wins 198 draws because the experimental hub adds burden and the connector adds cost. In this model, a regime lies on the four-axis Pareto frontier when no alternative is at least as good on exposure severity, public cost, the modelled voluntary-communication gap, and accuracy and strictly better on at least one of those criteria. Stylised baseline is on that frontier in all 200 draws at every horizon; the hub appears in 183, 180, 180, and 184 draws; and the connector appears in 164, 174, 200, and 200 draws at 120, 365, 1,000, and 2,000 days. A high Pareto frequency means that a regime is not strictly dominated on the selected criteria, not that it is preferred under all values or that it dominates the other regimes.

5.2 Pre-screening: lower modelled overstatement, larger modelled voluntary-communication gap

The hub reduces original material overstatement relative to the stylised baseline in 91.0% of 120-day draws, 88.5% at 365 days, 87.5% at 1,000 days, and 88.0% at 2,000 days. It reduces exposure-weighted severity in 85.0%, 90.5%, 89.0%, and 87.5% of draws. Mean paired improvements in the original material count are 8.65, 13.74, 24.20, and 45.99 claims as horizons lengthen. Because claim volume and exposure accumulate over time, these absolute effects should not be compared across horizons as rates.

Table 6. Principal paired differences relative to the stylised ex-post baseline

The two Share columns report the proportion of sampled configurations with an estimated improvement; they are sampled-configuration shares, not success probabilities. Negative Δ values for the modelled voluntary-communication gap and public cost indicate a larger gap or higher cost. Costs are simulation-scale EUR.

Days	Arm	Share: original count	Share: exposure severity	Mean Δ modelled voluntary-communication gap	Mean Δ public cost
120	Pre-screening hub	91.0%	85.0%	-0.00462	-EUR 24,290
120	Data connector	50.5%	40.5%	-0.000001	-EUR 64,071
365	Pre-screening hub	88.5%	90.5%	-0.00637	-EUR 26,462
365	Data connector	68.5%	60.5%	-0.00013	-EUR 70,352
1,000	Pre-screening hub	87.5%	89.0%	-0.00574	-EUR 28,751
1,000	Data connector	100.0%	97.0%	-0.00035	-EUR 86,192
2,000	Pre-screening hub	88.0%	87.5%	-0.00502	-EUR 32,088
2,000	Data connector	100.0%	100.0%	-0.00044	-EUR 111,608

The model does not justify calling every prevented publication fraud prevention. The hub can prompt revision of good-faith error, delay publication, or induce withdrawal. A defensible model-conditional interpretation is a mixture of revision and claim suppression. Because withdrawal mechanically adds one-seventh of the truthful benchmark to the modelled voluntary-communication gap measure, part of the headline relationship is encoded by the metric and requires alternative-definition and ablation analysis. Uptake is type-dependent and stochastic, not a strategic participation choice.

The trade-off is visible in the modelled voluntary-communication gap. The normalised paired effect is negative at every horizon: mean changes of -0.00462, -0.00637, -0.00574, and -0.00502 indicate a larger modelled voluntary-communication gap relative to the stylised baseline. The hub reduces this gap in only 0%, 3.0%, 4.5%, and 6.5% of draws. It also increases discounted public cost in every draw, with mean paired cost effects of -EUR 24,290, -EUR 26,462, -EUR 28,751, and -EUR 32,088. These are simulation-scale euros and should not be mapped to an EU budget.

Sensitivity results reinforce the mechanism. At 120 days, the largest partial rank correlations with the hub's severity improvement are participation scale (PRCC 0.531) and strictness (0.446); at 365 days they are 0.552 and 0.494. The same parameters have negative associations with the modelled voluntary-communication-gap effect: at 365 days, participation PRCC is -0.550 and strictness is -0.496. More participation and stricter screening improve modelled claim quality while worsening communication suppression. This is a structural trade-off within the model, not an externally validated elasticity.

5.3 Connector: horizon-dependent benefits and persistent cost

The archived connector pattern differs by horizon. Exposure-weighted severity has the favourable estimated sign in 40.5%, 60.5%, 97.0%, and 100.0% of sampled configurations at 120, 365, 1,000, and 2,000 days, respectively; the corresponding mean paired differences are −0.63, 44.61, 1,855.23, and 12,216.10 accumulated exposure-severity units. Original material-overstatement counts have the favourable sign in 50.5%, 68.5%, 100.0%, and 100.0% of configurations. These accumulated quantities are not rates and should not be compared across horizons as effect magnitudes.

The full-space long-horizon pattern is stronger than the earlier post hoc subset pattern, but it remains a model result under sampled ranges. It is consistent with recurring transfers, provenance, comparison, correction, and population detection. The design cannot attribute the pattern to those mechanisms, however, because horizon also changes reporting-cycle count, discounting, legal-calendar state, and market ticks. Nor does it demonstrate real-world effectiveness or cost-benefit.

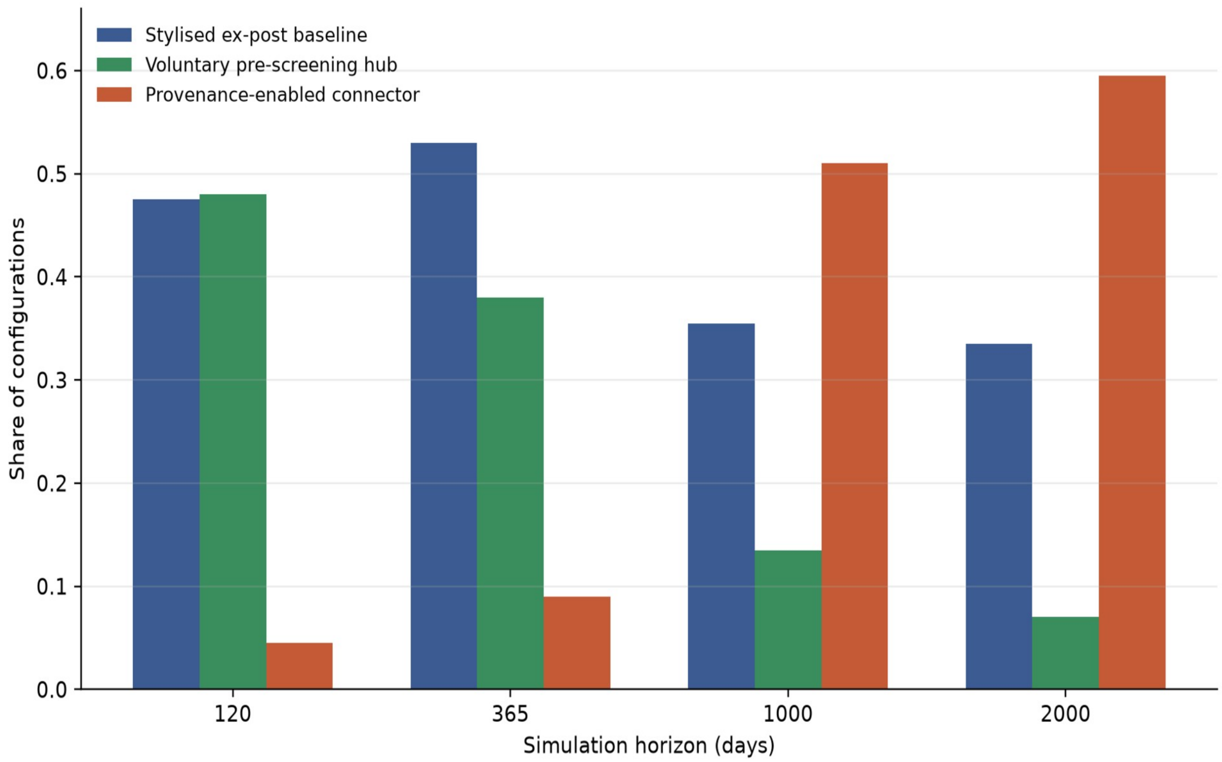

Figure 2. Share of configurations selecting each arm under default weights, by horizon.

Source: the four per-horizon draw-effect CSV files in results/summaries/ (archived campaign code identifier 1645f3589408503894c7e8f44a92ffe382a9a5b4+dirty). Each horizon contains 200 configurations and three paired replications.

The cost result remains adverse. The connector increases discounted public cost in every draw, with mean normalised effects of -EUR 64,071, -EUR 70,352, -EUR 86,192, and -EUR 111,608. Social discount rate is the strongest parameter associated with the connector's cost effect, with PRCC values of 0.756, 0.688, 0.716, and 0.839 across the four horizons. The connector has a near-zero mean effect on the modelled voluntary-communication gap, but the share with a favourable sign remains only 12.0%, 12.0%, 13.0%, and 10.5%. This should be interpreted as little systematic reduction in the modelled voluntary-communication gap, not as evidence of large communication suppression.

Connector severity effects become more sensitive to economy-wide behavioural settings over time. At 1,000 and 2,000 days, regulatory strictness has PRCC values of -0.616 and -0.647 with connector severity improvement, while compliance burden has -0.490 and -0.530. The result reflects interactions between baseline communication choices and the marginal information supplied by the connector. These coefficients are associational sensitivity diagnostics.

5.4 Conflict desk and source errors

Connector data create modelled disputes that the baseline encounters less often. Averaged over the 600 regime runs per horizon, the connector opens 0.87 conflict cases at 120 days, 2.28 at 365 days, 4.50 at 1,000 days, and 7.99 at 2,000 days. It consumes 0.71, 1.86, 3.87, and 6.88 investigation slots, respectively. At 365 days, neither the stylised baseline nor the hub opens a conflict in the campaign mean; at 1,000 days their means are 0.71 and 0.79 cases, compared with 4.50 for the connector. At 2,000 days they open 2.08 and 2.10, compared with 7.99.

Resolution is slower under the connector because modelled corrections and re-verification consume time. Mean connector resolution delay is 44.43 days at 365 days, 75.98 days at 1,000 days, and 102.47 days at 2,000 days. Relative to baseline, the normalised delay effects are -44.43, -58.27, and -46.64 days. At 120 days, open disputes have generally not matured into resolved-delay observations. The capacity reserve prevents indefinite starvation but competes with ordinary enforcement. Higher source error loads therefore burden the dispute channel.

The safeguard is that source error does not become a direct sanction channel. A register mismatch produces a conflict; a material case can escalate only after independent corroboration. At 2,000 days, connector runs average 1.18 corroborated escalations and 2.79 register corrections, compared with 0.36 escalations and 1.40 register corrections under the stylised baseline. These figures show procedural activity, not evidence of real-world error rates. The stress tests described in the repository use high register-error scenarios to verify that unresolved conflict cannot support a monetary sanction.

5.5 Claim-quality–modelled voluntary-communication-gap frontier and real-economy interpretation

The simulated policy problem is a frontier rather than a one-dimensional detection contest. Draws with lower exposure-weighted severity can have a larger modelled voluntary-communication gap, and draws with extensive truthful communication can expose more claims to delayed enforcement. Figure 3 plots this frontier for the 120-day campaign.

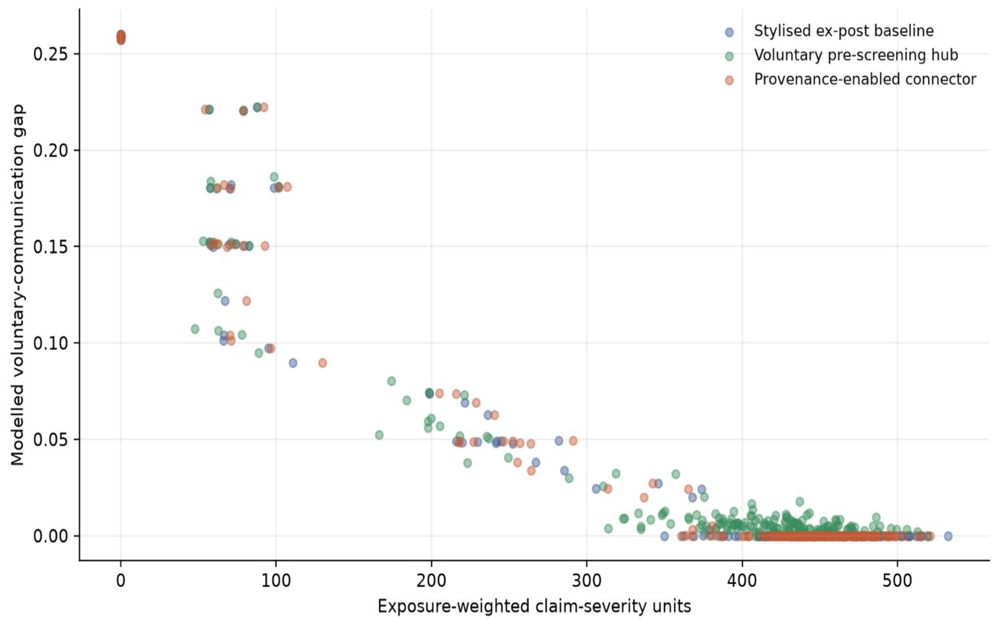

Figure 3. Exposure-weighted claim severity and modelled voluntary-communication gap at 120 days.

Source: results/summaries/horizon_120d_draw_effects.csv (archived campaign code identifier 1645f3589408503894c7e8f44a92ffe382a9a5b4+dirty). Both axes are researcher-defined model quantities.

The real-economy mappings give the modelled communication outcomes downstream consequences for demand, valuation, trust, productivity, and turnover. Their directions and magnitudes are uncalibrated stylisations, not observed causal effects. Investor-signal distortion is therefore an internal response metric. A limited own-price-to-balance financing path exists, and its materiality is unknown until dynamically ablated.

5.6 Summary of consistent and fragile model findings

Table 7. Interpretive stability

Finding	Evidence in archived campaign	Permissible interpretation
Hub lowers claim harm	Original and severity improve in about 85-91% of draws across horizons	Consistent across archived sampled configurations; mechanism includes revision and withdrawal
Hub increases the modelled voluntary-communication gap	The modelled voluntary-communication gap decreases in only 0–6.5% of draws; public cost worsens in all draws	Prevention has a communication-suppression and fiscal trade-off
Connector is horizon dependent	Severity improves in 40.5%, 60.5%, 97%, and 100% of draws	Short-horizon null-to-mixed pattern; long-horizon pattern is calendar/cadence-confounded
Connector is costly	Discounted public cost worsens in all draws at every horizon	Long-run claim-quality gains do not imply simulated cost savings
Rankings are unstable	80.5% cross-horizon reversal; 45.5-70% weight-sensitive within horizon	No universal winner or unconditional policy recommendation
Conflict safeguards matter	Connector opens and investigates more disputes; escalation requires corroboration	Source errors load procedure rather than directly producing sanctions

Across the archived three-replication design, the pre-screening arm has a favourable estimated claim-quality sign in a large majority of sampled configurations while increasing the modelled voluntary-communication gap and public cost. The connector pattern varies strongly by horizon and is cost-intensive. These statements remain exploratory because winner instability, metric construction, calendar confounding, unsampled constants, and limited replication constrain interpretation.

5.7 Separate financial-market stylised-fact diagnostics

A separate fixed-parameter diagnostic campaign contains 30 independently seeded 10,000-day paths, each discarding a 2,000-day burn-in and analysing the primary listing. It uses 30 traders, whereas the policy campaign uses eight; credit and manipulators are disabled. It therefore describes a separate market configuration and does not validate the policy-run configuration. The burn-in lacks a reported convergence sensitivity, and the aggregate manifest contains a stale output-root field. Seed-level results are retained as illustrative stylised-fact diagnostics.

Table 8. Separate market-configuration stylised-fact diagnostics (30 independent seeds)

Source: results/financial_validation/pilot_30seeds/diagnostics/. The seed is the unit of inference. R/P/N = seeds classified reproduced / partially reproduced / not reproduced.

Fact	Cross-seed diagnostic (means over 30 seeds)	R/P/N	Status
Volatility clustering	Absolute-return ACF 0.170 and squared-return ACF 0.183 at lag 1, beyond the 95% band at multiple lags	30/0/0	Reproduced
Positive spread and depth	Mean spread 1.38 ticks; two-sided depth in essentially all snapshots	30/0/0	Reproduced
Trade-sign persistence	Lag-1 trade-sign ACF 0.486 against an approximate 0.009 white-noise band	30/0/0	Reproduced
Cancellation activity	Cancellation-to-submission ratio 0.537	30/0/0	Reproduced
Volume-volatility relation	Mean daily volume-volatility correlation 0.113	17/13/0	Partially reproduced
Fat-tailed returns	Mean excess kurtosis 1.99 (cross-seed SD 1.33); mean Hill alpha 5.38	3/27/0	Partially reproduced
Weak linear return autocorrelation	Mean lag-1 return ACF -0.082, small but outside the white-noise band	0/29/1	Partially reproduced
Positive price impact	Mean correlation of log volume with immediate absolute mid-price movement -0.013 (95% half-width 0.004)	0/4/26	Not reproduced

Four implemented microstructure/volatility criteria are met in every diagnostic seed. Return-distribution criteria are mixed: excess kurtosis is positive but exceeds 3 in only a minority of seeds, the mean Hill estimate is 5.4, and short-lag return autocorrelation is small but negative beyond the implemented white-noise band. These are results under the repository's stated thresholds; no uncited empirical tail benchmark is used here.

The repository's positive-impact criterion is not met. Its unsigned immediate-horizon statistic correlates log trade volume with absolute mid-price movement; the cross-seed mean is −0.013 with a reported half-width of 0.004, and 26 of 30 seeds are classified as not reproduced. This statistic is not a signed multi-horizon price-impact event study and should not support broader empirical claims.

These diagnostics bound interpretation. Market-side quantities are internal model responses; neither the market layer nor the policy configuration is empirically validated. Because a limited treasury-price-to-balance pathway exists, the materiality of price-mediated financing for policy rankings is unknown until a dynamic ablation is performed.

6. Discussion

This study integrates actor-specific information constraints, capacity-limited procedure, voluntary communication, and an endogenous market context in one exploratory simulation. The archived outputs show different modelled trade-offs for early rule-based review and recurrent data exchange. They do not establish novelty priority, legal feasibility, empirical effectiveness, or welfare optimality. The literature was reviewed and the cited sources were manually verified by the author for existence, relevance, and consistency with the study's scope.

The arms are comparable only as model mechanisms. The baseline is a stylised collection of ex-post processes; the hub and connector are different add-ons with different eligible populations, reach, opportunities, delays, and structurally additional costs. Because the design imposes no common budget or staff constraint and contains no combined hub-plus-connector arm, it cannot establish legal superiority, cost-effectiveness, or complementarity. Equal-resource, equal-reach, and factorial comparisons are required.

The long-horizon connector pattern is consistent with accumulated authorisation, transfer, reconciliation, and correction, but horizon also changes reporting cycles, discounting, and the legal-calendar state because all runs start on 1 January 2026. Legal-calendar freezes, start-date variation, cadence sensitivity, and mechanism ablations are needed before the accumulation interpretation can be identified. Real deployment of either add-on would additionally require institutional mandate, due process, human oversight, contestability, data-access authority, trade-secret safeguards, data protection, cybersecurity, procurement, and review.

The information-safe procedural design is essential to that comparison. Supervisors and assurance providers receive uncertain records rather than latent truth; cases proceed through screening, evidence requests, correction, investigation, decision, publication, and closure; and conflicts consume capacity without supporting sanctions unless evidence is corroborated. This makes false positives, inconclusive cases, exposure duration, and modelled voluntary-communication-gap outcomes of institutional procedure rather than assumptions away from the analysis.

The paired global design then tests these mechanisms across uncertain environments and normative choices. Raw effects, Pareto membership, rank frequency, parameter sensitivity, and alternative weights constrain interpretation of the composite score. The LEGAL, LEGAL-ANCHOR, EXPERIMENT, and STYLIZATION registry makes the source and role of assumptions visible without implying empirical calibration.

7. Limitations

Seven limitations bound the interpretation of the experiment.

First, the model is neither empirically calibrated nor a source of causal inference. Enforcement frequencies, claim prevalence, behavioural responses, emissions, administrative costs, and downstream mappings are not estimated from observed data. The LHS examines a selected set of chosen ranges; paired initial seeds identify only within-model differences. Uptake is fixed-profile-dependent stochastic participation rather than strategic endogenous selection. Outputs are not real-world forecasts.

Second, legal and institutional implementation is reduced-form and scenario-specific. National transposition, jurisdiction, appeals, remedies, enforcement cooperation, complete CSDDD undertaking scope, the full ESRS datapoint architecture, Taxonomy technical criteria, lifecycle assessment, assurance standards, and European Green Bond workflows require separate legal and technical treatment. The connector also abstracts from the engineering, governance, lawful-basis, cybersecurity, and organisational requirements of the GDPR and NIS 2 Directive (European Parliament and Council, 2016b; 2022b). The paper is not legal advice.

Third, measurement interoperability and environmental coverage are partial. The model detects unit mismatch but does not represent conversion evidence; Scope 3, facilities, subsidiaries, individual machines, service-production boundaries, and supply chains are incomplete. Methodology changes, boundary differences, staleness, and missing coverage enter through stylised uncertainty. The detector also cannot resolve production-driven absolute reductions, intensity-versus-absolute framing without production evidence, coordinated narratives, or audience perceptions of technically true claims. The connector reconciles available records but cannot create missing meters, supplier data, or a complete environmental truth system. Claim-quality outcomes must therefore be distinguished from physical-emissions outcomes.

Fourth, the composite score and its weights are normative EXPERIMENT aggregations, not an absolute welfare function. Default winner shares describe one value system; alternative weights and frequent Pareto co-membership show that rankings depend on accuracy, the modelled voluntary-communication gap, cost, participation, and burden priorities. The exogenous EUR 1,000 consumer budget, simulation-scale corporate balances, and simulation euros are accounting devices and cannot be read as EU fiscal costs or firm forecasts.

Fifth, statistical and computational reproducibility are limited. The authoritative campaign uses three paired replications per draw, so rare events and draw-level rankings can remain noisy. The executed 15-replication extension on 12 post hoc outcome-diverse configurations narrows paired intervals in 92.3% of comparable cells yet confirms the three-replication default-weight winner in only 32 of 48 draw-horizon cells; distributional effects, improvement fractions, and Pareto membership are therefore more reliable than draw-level league tables. Partial rank correlations and standardised coefficients can miss non-monotonicity and interactions. In addition, the shipped campaign manifests record a git hash suffixed +dirty. The archival provenance package preserves the authoritative outputs, configuration, seed schedules, recoverable environment information, and SHA-256 checksums, but the original dirty diff was not retained. A later direct-child commit is only a partially reconstructed candidate source snapshot; recovery of the original dirty worktree and bit-level reproducibility are not claimed.

Sixth, the separate financial-market diagnostics do not validate the eight-trader policy configuration, and a limited own-price-mediated treasury-financing pathway enters corporate balance-dependent choices. Across 30 independent seeds, volatility clustering, positive spread and depth, trade-sign persistence, and cancellation activity are reproduced; fat tails, the volume-volatility relation, and weak linear return autocorrelation are only partially reproduced; and positive price impact is not reproduced under the implemented immediate-impact diagnostic. Market-side quantities are therefore internal responses, and no policy ranking is evidence of equity-price discipline over corporate environmental behaviour.

Seventh, temporal and design choices constrain interpretation. Simulated calendar days are also market ticks; a fixed 60-day reporting cadence, fixed firm profiles, and selected unsampled constants shape exposure and workload. Longer horizons activate different legal-calendar states. The 28-dimensional LHS covers a selected subset of model uncertainty, and integer-valued inputs must be analysed at their applied rounded values.

8. Conclusion

Within the experiment, preventive pre-screening frequently lowers material overstatement and exposure-weighted severity because it acts before publication, but recurrent review and revision costs, selective participation, delay, and withdrawal increase the modelled voluntary-communication gap and reduce its default-weight competitiveness over longer horizons. The hub therefore continues to improve claim quality in a large share of draws; it does not simply fail.

The connector shows a mixed short-horizon and stronger long-horizon pattern. That pattern is consistent with recurring evidence and correction, but the current design cannot distinguish accumulation from reporting-cycle, discounting, legal-calendar, or market-tick effects. The connector is the most frequent arm under default weights at 1,000 and 2,000 days, while remaining costly, generating conflicts, and never Pareto-dominating the stylised baseline. The 80.5% cross-horizon reversal rate supports only a conditional comparison under the specified design.

The model should be read as version zero of a broader research programme, not as a policy recommendation engine, empirical forecast, legal evaluation, or causal study. Future versions should replace the current ten fixed corporate profiles with a larger, heterogeneous population of firm agents whose communication, investment, and compliance choices can evolve through opportunism, adherence to rules, conviction, learning, and interaction with stakeholders. Alongside this extension, future research should calibrate behavioural, enforcement, measurement, and cost channels; extend facility and value-chain representation; increase replication; test business-day and reporting-cadence conventions; isolate legal-calendar and accumulation mechanisms; and develop jurisdiction-specific legal modules. Financial work should repair the price-impact diagnostic and dynamically ablate the existing treasury-price-to-balance pathway before any market-discipline claim.

Data and Code Availability

The authoritative publication results are preserved in the project archive as machine-readable configuration files, manifests, run-level outputs, summaries, figures, and logs. The 846 files constituting the archived global-LHS campaign have been verified against their recorded SHA-256 checksums; the archived results therefore permit exact inspection and re-aggregation of the results reported in this paper.

The campaign was executed from source identified in its manifests as `1645f3589408503894c7e8f44a92ffe382a9a5b4+dirty`. The original dirty working tree was not retained and cannot be recovered as an authenticated source snapshot. The best available reconstruction is commit `00134a694e653fc2ce74b62eab1e8c6e9d0f67f7`, a direct child of the recorded base commit, created shortly after completion of the campaign and containing the relevant source files and archived outputs. It is provided as a frozen candidate source snapshot, not as a verified copy of the original dirty worktree.

The archive therefore supports exact verification of the published outputs and a partially reconstructed source provenance, but it does not claim bitwise executable replication of the historical campaign. A complete executable replication release would require a clean tagged source tree, a locked execution environment, and a fresh run validated against the archived output checksums.

The code, archived outputs, provenance materials, and candidate source snapshot are available at:

https://github.com/francescoromano1407-stack/Financial-market-simulation

https://github.com/francescoromano1407-stack/Financial-market-simulation/commit/00134a694e653fc2ce74b62eab1e8c6e9d0f67f7

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Conflicts of Interest

The author declares no conflicts of interest.

Ethics Statement

No human participants, personal data, or animal subjects were involved in this research. Ethical approval was therefore not required.

Author Contributions

Francesco Romano: Conceptualization; Methodology; Software; Validation; Formal Analysis; Investigation; Data Curation; Writing – Original Draft; Writing – Review & Editing; Visualization; Project Administration. The author is solely responsible for all aspects of the work.

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Appendix A. Parameter Classification and Selected Ranges

The registry defines four role classes. LEGAL values implement binding dates, conjunctive thresholds, or track-specific ceilings. LEGAL-ANCHOR values use a legal boundary to define an experimental target population but do not create an exemption. EXPERIMENT values specify institutional mechanisms or evaluation choices. STYLIZATION values encode behavioural, accounting, or scale relationships. Evidence classes separately identify legally mandated, reference-class, or illustrative-scenario values.

Table 9. Parameter-role taxonomy and sampled families

Detailed defaults, bounds, code locations, and justifications appear in docs/PARAMETER_REGISTRY.md.

Family	Examples	Role class	Selected LHS ranges
Legal calendar	Application dates, CSRD thresholds, gated ceilings	LEGAL	Fixed; not sampled
Target eligibility	Hub 1,000-employee line	LEGAL-ANCHOR	Fixed; not sampled
Sanctions/capacity	Benefit factor, evidence and investigation slots, surveillance	EXPERIMENT / STYLIZATION	0.5-3x benefit; 5-40 requests; 1-12 investigations
Conflict desk	Reserve share, resolution days	EXPERIMENT	0-0.60 reserve; 5-60 days
Hub	Participation, strictness, noise, delay	EXPERIMENT	0.2-1.5; 0.1-0.95; 0-0.50; 0-20 days
Connector	Coverage, mismatch, register error, staleness, downtime	EXPERIMENT	Coverage 0.5-1.2; failure probabilities 0-0.25
Behaviour	Consumer, investor, workforce, compliance burden	STYLIZATION	Documented registry ranges; no empirical calibration
Evaluation	Social discount rate, score weights	EXPERIMENT	0-7% per year; five discrete weight scenarios

All 28 globally sampled dimensions are EXPERIMENT or STYLIZATION. Examples include evidence-request capacity of 5-40 cases per period; investigation capacity of 1-12; random surveillance of 0-0.30; correction windows of 10-90 days; hub strictness of 0.10-0.95 and noise of 0-0.50; connector mismatch of 0-0.10, register error of 0-0.05, staleness of 0-0.25, downtime of 0-0.10, and correction delay of 10-120 days; consumer preference scale of 0.5-1.5; investor sophistication of 0.05-0.80; workforce trust loss of 0.10-1.00; compliance burden scale of 0.25-2.50; and social discount rate of 0-0.07 per year. These ranges define the computational experiment and are not confidence intervals around estimated parameters.

Appendix B. Outcome Definitions

Original material overstatements. Count of claims whose original published value exceeds ex-post truth by at least the evaluator's experimental materiality threshold. This is research-only.

Live-uncorrected material overstatements. Material claims still active after prospective correction and withdrawal events.

Exposure-weighted severity. Claim severity multiplied by exposure days, preserving harm from delayed correction. A discounted counterpart is used in comparisons while the undiscounted ledger is retained.

Population detection recall. Share of all original material claims that receive a relevant detected outcome. Unlike screening-conditioned recall, the denominator includes claims never selected because of limited capacity. This is research-only.

Modelled voluntary-communication gap. Positive difference between the truthful communication benchmark and chosen voluntary communication intensity, averaged over relevant observations. This is a constructed model measure and does not establish observed behaviour or managerial intent.

Meaningful pre-publication revision. Revision that addresses a model-classified material hub issue. Informational or noise flags do not count as prevention.

Conflict metrics. Cases opened, investigations consuming capacity, unresolved cases, resolution delay, firm confirmation, register correction, claim correction, dismissal, and corroborated escalation.

Discounted total public cost. State policy and regulator time cost discounted at the draw-specific social rate. It is a simulation-scale accounting metric.

Pareto membership. Non-dominance on lower severity, lower public cost, a smaller modelled voluntary-communication gap, and higher accuracy. It is reported alongside, not replaced by, composite ranks.

Appendix C. Reproducibility and Seed Strategy

The archived configuration and provenance records document the campaign design and seed schedule. Because the exact historical dirty source tree was not preserved, the original campaign is not claimed to be bitwise reproducible; the preserved outputs can be re-aggregated, and the current code can be evaluated subject to the provenance limitations described in the Data and Code Availability section.

The top-level publication driver refuses fewer than 200 draws, three paired replications, or 120 days unless explicitly placed in development mode. Re-aggregation can be performed from the preserved outputs without re-running simulations. The archived configuration specifies run_global_lhs_campaign.py --full-robustness. All outputs should be interpreted with the corresponding machine-readable manifest and archival provenance package, which records the original +dirty label, the partially reconstructed candidate source snapshot, and the limits on reproducing the original working tree.

Appendix D. Additional Robustness Interpretation

Across all horizons, the communication-protection and cost-focused scenarios favour the stylised baseline more strongly than the default scenario. At 2,000 days, the stylised baseline remains the modal choice under both cost-focused and burden-protective weights, even though the connector has the highest default winner share. These distinct robustness comparisons show that the long-horizon default ordering remains value-dependent; they do not establish convergence to a universal policy.